
12 Research Methodology

New Description

Outcome	Weight
Explore of IT Areas	◆◆◆◆

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Outcome	Weight
Context of Research Project	◆◆◆◆

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Outcome	Weight
Communication of Research Outcomes	◆◆◆◆

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Outcome	Weight
Validation of Objectives and Methodology	◆◆◆◆

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Outcome	Weight
Application of Skills and Knowledge	◆◆◆◆

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DEAKIN UNIVERSITY

RESEARCH TRAINING AND PROJECT

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Research Methodology

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September 9, 2025



Research Methodology for Detection and Simulation of GPS Spoofing in Mobile Robots Using Sensor Cross-Validation

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SIT723 Research Training and Project

Deakin University

September 9, 2025

Abstract

This research methodology document presents my comprehensive approach to investigating GPS spoofing detection in autonomous mobile robots through data-driven machine learning techniques. I have developed a structured research plan that integrates LSTM networks, hybrid sensor fusion with Dynamic Time Warping, and multi-layered signal analysis within both simulation and hardware implementation frameworks. My methodology addresses three core research questions through systematic experimental design, emphasizing the critical relationship between simulated validation and real-world hardware deployment. This document outlines my detailed timeline, ethical considerations, milestone tracking, and sustainability measures for conducting high-quality, reproducible research in cybersecurity for autonomous systems.

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1 Introduction

Through my extensive literature review and methodological analysis, I have identified a critical need for robust GPS spoofing detection mechanisms in autonomous mobile robots. My research methodology is founded on data-driven machine learning approaches that bridge the gap between simulation-based validation and practical hardware implementation. I recognize that successful spoofing detection requires sophisticated methodologies combining temporal analysis, sensor integration, and multi-layered verification systems to address the evolving threat landscape in autonomous vehicle security.

My approach emphasizes the fundamental relationship between simulation environments and hardware deployment, understanding that simulation provides the controlled environment necessary for algorithm development while hardware implementation validates real-world applicability. I have structured my methodology to leverage both paradigms synergistically, using simulation for systematic algorithm development and hardware platforms for validation under realistic operational conditions.

The research problem I address centers on developing effective methodologies for real-time GPS spoofing detection that can operate within the computational constraints of mobile robotic platforms while maintaining high detection accuracy across diverse attack scenarios. My investigation acknowledges that no single detection method can adequately address all types of spoofing attacks while maintaining the computational efficiency required for real-time autonomous operations.

2 Research Questions

Building upon my comprehensive literature analysis, I have formulated three interconnected research questions that guide my methodological approach:

2.1 Primary Research Question (RQ1)

What are the primary vulnerabilities of GPS in autonomous mobile robots to spoofing attacks?

Rationale: My investigation into current literature reveals fundamental weaknesses in civilian GPS signals, but I have identified a gap in comprehensive analysis specific to mobile robotic platforms. Understanding these vulnerabilities is essential for developing targeted detection mechanisms that address platform-specific constraints and operational requirements.

2.2 Secondary Research Question (RQ2)

How can sensor fusion techniques effectively detect GPS spoofing in real-time using data-driven machine learning approaches?

Rationale: My analysis shows that existing sensor fusion approaches demonstrate promise but lack implementation specifics for real-time operation on embedded platforms. I aim to develop cross-validation methodologies between GPS, IMU, and odometry sensors optimized for mobile robots through machine learning techniques.

2.3 Tertiary Research Question (RQ3)

How can a dynamic GPS Trust Index be designed using machine learning to quantify the reliability of GPS data continuously?

Rationale: Current detection methods provide binary outputs (spoofed/authentic) but lack quantitative reliability assessment. I propose developing a machine learning-based dynamic trust

index that enables gradual response to varying threat levels and provides continuous assessment of GPS data integrity.

3 Research Methods Overview

Through my comprehensive analysis of available methodologies, I have selected three complementary research approaches that demonstrate the greatest potential for effective GPS spoofing detection in autonomous mobile robots. My selection criteria prioritize practical implementation feasibility, detection accuracy, computational efficiency, and adaptability to various attack scenarios.

3.1 Methodology Comparison and Selection

Table 1: Comparative Analysis of Research Methodologies for My Research

Method	Realism	Cost	Reproducibility	Stability	ML Integration
LSTM Networks	High	Medium	High	High	Excellent
Sensor Fusion + DTW	High	Medium	High	High	Good
Multi-layered Analysis	High	Medium	High	High	Excellent
Signal Processing	Medium	Low	High	High	Limited
Hardware Testing	Very High	High	Low	Low	Variable

3.2 Selected Research Methods

3.2.1 Method 1: LSTM Network-Based Detection

I have selected Long Short-Term Memory (LSTM) networks as my primary machine learning approach due to their demonstrated effectiveness in temporal pattern recognition and vehicle state prediction. My implementation will focus on training LSTM networks on historical vehicle movement data to establish prediction models for normal navigation behavior, enabling detection of sophisticated spoofing attacks that evolve gradually over time.

3.2.2 Method 2: Hybrid Sensor Fusion with Dynamic Time Warping

My second methodology combines sensor fusion techniques with Dynamic Time Warping (DTW) algorithms for enhanced detection capabilities. I will integrate accelerometer, gyroscope, steering angle, and speedometer data through machine learning-enhanced fusion algorithms to create comprehensive situational awareness that exceeds single-sensor capabilities.

3.2.3 Method 3: Multi-layered Signal and Behavioral Analysis

My third approach implements multi-layered analysis frameworks that combine signal characteristic monitoring with behavioral pattern analysis through machine learning classification. I will develop adaptive machine learning models that can distinguish between authentic and spoofed signals while maintaining high accuracy across diverse operational scenarios.

4 Implementation Stack Architecture

I have designed a comprehensive implementation stack that demonstrates the relationship between simulation and hardware deployment:

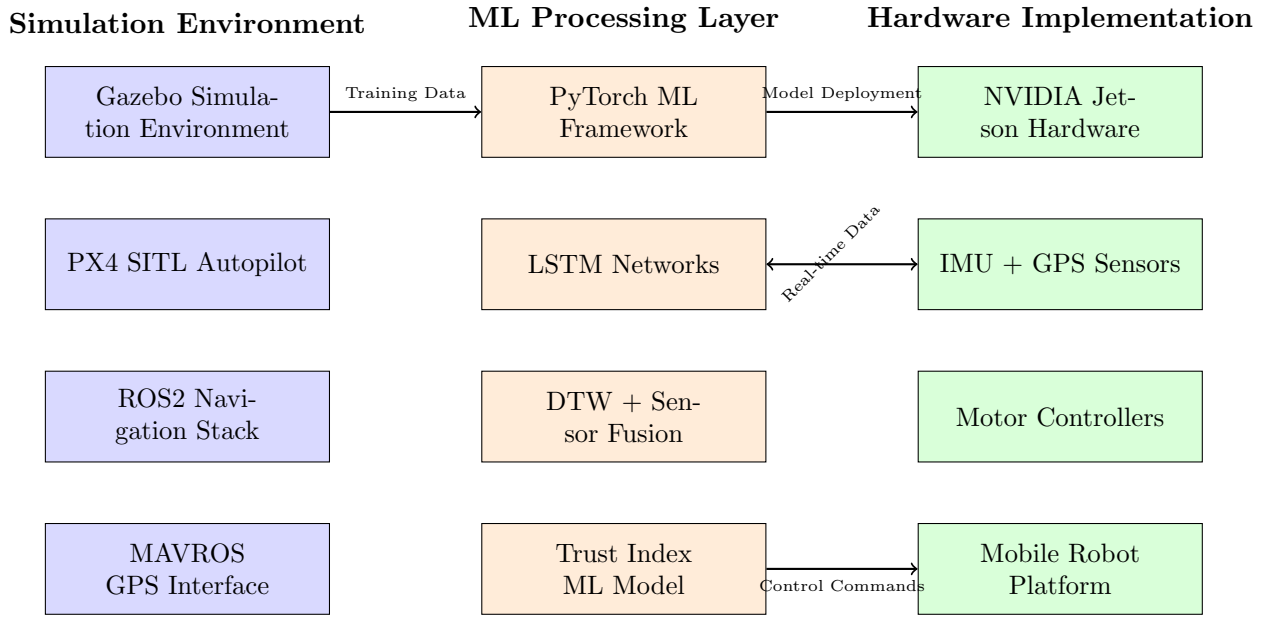


Figure 1: My Implementation Stack Architecture: Simulation to Hardware Pipeline

4.1 Simulation-Hardware Integration Strategy

My approach recognizes that simulation environments provide essential controlled conditions for algorithm development, while hardware implementation validates real-world performance. I have structured my methodology to leverage this relationship through:

Simulation Phase: I will use Gazebo with PX4 SITL to create controlled spoofing scenarios, enabling systematic evaluation of machine learning algorithms across diverse attack types. This environment allows me to generate large datasets necessary for training deep learning models while maintaining complete control over experimental conditions.

Hardware Validation Phase: I will deploy trained models on NVIDIA Jetson hardware integrated with actual GPS receivers and IMU sensors. This phase validates that simulation-trained models maintain performance under real-world conditions with actual sensor noise, environmental interference, and computational constraints.

5 Research Plan and Timeline

My research plan follows a systematic progression from theoretical development through simulation validation to hardware implementation. I have structured the timeline to allow for iterative refinement and validation at each stage.

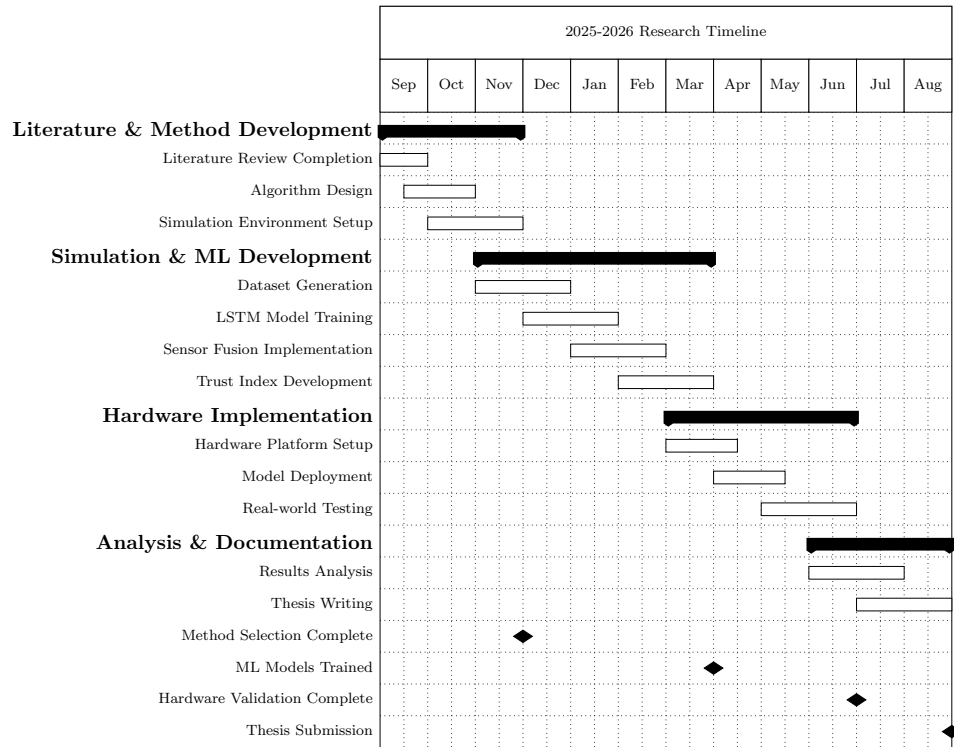


Figure 2: My Research Timeline and Milestones

5.1 Detailed Phase Breakdown

5.1.1 Phase 1: Literature Review and Method Development (Sep-Nov 2025)

During this phase, I will complete my literature review and finalize algorithm designs. I will establish my simulation environment using Docker containers running PX4 SITL with Gazebo, ensuring reproducible experimental conditions.

5.1.2 Phase 2: Simulation and ML Development (Nov 2025 - Mar 2026)

I will focus on generating comprehensive training datasets through controlled spoofing scenarios in simulation. My LSTM networks will be trained on vehicle movement patterns, while sensor fusion algorithms will be developed and validated through systematic testing.

5.1.3 Phase 3: Hardware Implementation (Feb-Jun 2026)

I will deploy my trained models on NVIDIA Jetson hardware integrated with GPS receivers and IMU sensors. This phase validates simulation results under real-world conditions and assesses computational performance on embedded platforms.

5.1.4 Phase 4: Analysis and Documentation (May-Aug 2026)

I will conduct comprehensive analysis of results from both simulation and hardware testing phases. My thesis will be written during this period, documenting methodologies, results, and contributions to the field.

6 Data-Driven Machine Learning Approach

My research emphasizes data-driven methodologies that leverage machine learning techniques for robust spoofing detection. I have structured my approach around three key machine learning

components:

6.1 LSTM Network Architecture

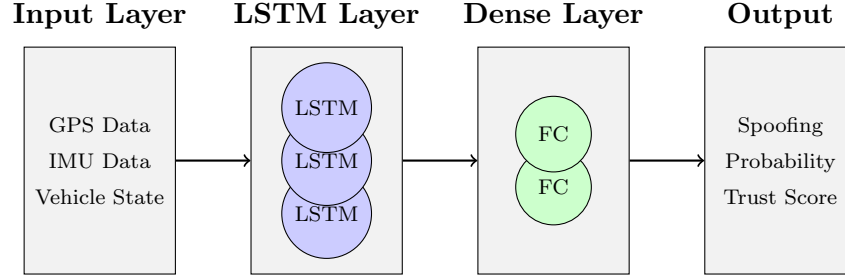


Figure 3: My LSTM Network Architecture for Temporal Pattern Analysis

My LSTM implementation processes sequential sensor data to identify temporal patterns characteristic of authentic GPS navigation versus spoofing attacks. I will train these networks on comprehensive datasets generated through simulation, enabling detection of gradual drift attacks that evolve over time.

6.2 Sensor Fusion ML Framework

I have designed a machine learning-enhanced sensor fusion framework that processes data from multiple sensors simultaneously:

Algorithm 1 My Data-Driven Sensor Fusion Algorithm

- 1: **Input:** GPS data G_t , IMU data I_t , Vehicle state V_t
 - 2: **Output:** Trust score T_t , Spoofing probability P_t
 - 3: $features \leftarrow extractFeatures(G_t, I_t, V_t)$
 - 4: $lstm_output \leftarrow LSTM(features)$
 - 5: $dtw_score \leftarrow DTW(V_t.steering, expected_steering)$
 - 6: $fusion_vector \leftarrow concatenate(lstm_output, dtw_score)$
 - 7: $P_t \leftarrow MLClassifier(fusion_vector)$
 - 8: $T_t \leftarrow calculateTrustIndex(P_t, historical_data)$
 - return** T_t, P_t
-

6.3 Dynamic Trust Index ML Model

My trust index model uses reinforcement learning principles to continuously adapt trust scores based on operational experience:

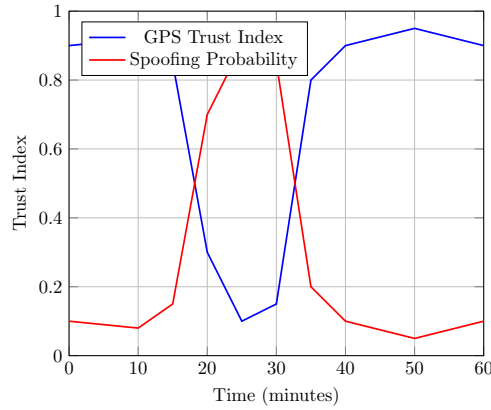


Figure 4: Dynamic Trust Index Response During Simulated Spoofing Attack

7 Ethical Considerations

My research addresses several critical ethical considerations inherent in GPS spoofing research:

7.1 Dual-Use Research Management

I acknowledge the dual-use nature of spoofing research and have structured my approach to focus exclusively on defensive applications. My research will avoid developing new attack methodologies and will not publish detailed specifications for spoofing equipment construction.

7.2 Safety and Controlled Testing

I will conduct all testing in controlled simulation environments to eliminate risks to public safety. My hardware validation will use low-power signals in shielded laboratory environments, with coordination with relevant regulatory bodies as required.

7.3 Data Privacy and Security

All data collected during my research will be anonymized and stored according to Deakin University's data governance policies. I will implement secure data handling procedures and ensure compliance with privacy regulations.

7.4 Responsible Disclosure

Any vulnerabilities discovered during my research will be reported through appropriate channels, following established responsible disclosure protocols. I will coordinate with relevant stakeholders before public disclosure of findings.

8 Milestones and Progress Tracking

My research includes specific milestones that mark critical achievements and enable progress monitoring:

Table 2: Key Research Milestones and Success Criteria

Milestone	Description	Success Criteria	Timeline
Simulation Environment	Complete PX4-Gazebo setup	Successful spoofing scenario generation	Nov 2025
ML Model Training	LSTM networks trained	≥95% accuracy on validation data	Mar 2026
Hardware Integration	Jetson platform deployment	Real-time processing <100ms latency	Jun 2026
Validation Complete	Real-world testing finished	Algorithm performance validated	Jul 2026

8.1 Risk Mitigation Strategies

I have identified potential risks and developed mitigation strategies:

Technical Risks: Hardware compatibility issues, algorithm performance degradation *Mitigation:* Early hardware testing, performance benchmarking, backup algorithm implementations

Timeline Risks: Delays in simulation setup, training data generation issues *Mitigation:* Parallel development tracks, contingency time allocation, simplified model alternatives

9 Sustainability and Impact

My research is designed to provide long-term value to the cybersecurity and autonomous systems communities:

9.1 Open Source Contribution

I will release all simulation code, trained models, and experimental frameworks as open-source contributions, enabling reproducible research and community collaboration.

9.2 Industry Application

My methodologies are designed for practical deployment on commercial autonomous platforms, with attention to computational constraints and real-world operational requirements.

9.3 Academic Dissemination

I plan to publish findings in peer-reviewed conferences and journals, contributing to the broader knowledge base in autonomous vehicle security.

9.4 Educational Impact

My work will contribute to educational resources at Deakin University, providing practical examples for future cybersecurity and robotics students.

10 Expected Contributions and Significance

Through my systematic approach, I expect to make several significant contributions:

10.1 Methodological Contributions

- Novel integration of LSTM networks with sensor fusion for real-time spoofing detection
- Data-driven trust index model that provides continuous reliability assessment
- Comprehensive simulation-to-hardware validation framework

10.2 Practical Contributions

- Open-source implementation suitable for deployment on resource-constrained platforms
- Reproducible experimental framework for GPS spoofing research
- Validated algorithms demonstrating effectiveness under real-world conditions

10.3 Scientific Contributions

- Enhanced understanding of GPS vulnerabilities specific to mobile robotic platforms
- Quantitative analysis of machine learning effectiveness for temporal spoofing detection
- Comparative evaluation of simulation versus hardware validation approaches

11 Conclusion

My research methodology provides a comprehensive framework for investigating GPS spoofing detection through data-driven machine learning approaches. The integration of LSTM networks, sensor fusion with Dynamic Time Warping, and multi-layered analysis creates a robust detection system suitable for autonomous mobile robots.

The systematic progression from simulation-based development to hardware validation ensures that my algorithms are both theoretically sound and practically deployable. My emphasis on open-source contribution and reproducible research practices will benefit the broader research community while addressing critical cybersecurity challenges in autonomous systems.

Through careful attention to ethical considerations, timeline management, and sustainability measures, my research is positioned to make significant contributions to the field while maintaining the highest standards of research integrity and practical applicability.

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